

Summary and Findings: Loan Approval Analysis

This study aimed to identify the key factors influencing loan approval decisions. The primary variables analyzed were **income, loan amount, credit score, and employment experience** to determine their impact on approval rates.

For this project, a **dataset from Kaggle containing 44,972 loan applications** was used. The data was pre-processed, with extreme outliers removed to improve the accuracy of the analysis. The following key findings were identified:

Descriptive Statistics & Data Distribution

- **All variables were positively skewed**, except for credit scores, which followed an approximately normal distribution.
 - **Age:** Most applicants were between **24-35 years old**.
 - **Income:** Right-skewed distribution, with most earning **below \$100,000**.
 - **Loan Amount:** Most loans ranged from **\$5,000-\$15,000**, with a few high-value outliers.
 - **Credit Score:** The majority of applicants had scores **above 600**.
- **PMF & CDF Analysis**
 - **Probability Mass Function (PMF)** showed a **higher approval rate for lower-income applicants**, which was contrary to expectations.
 - **Cumulative Distribution Function (CDF)** confirmed that most borrowers fell within common income and loan amount ranges.
- **Correlation & Hypothesis Testing**
 - **Loan Amount vs. Applicant Income:** Moderate positive correlation (**$r = 0.36$, $p < 0.0001$**), indicating that higher-income applicants tend to receive larger loans.
 - **Loan-to-Income Ratio & Loan Status:** A strong negative relationship—**higher loan percentages relative to income were associated with increased rejection probability ($p < 0.0001$)**.
 - **Employment Experience & Loan Approval:** **Extremely weak correlation ($r = -0.02$)**, despite statistical significance.
 - **Credit Score & Interest Rate:** **No meaningful relationship ($r = 0.01$, $p = 0.0176$)**, suggesting that other factors influence interest rates.

Challenges & Limitations

- **Most variables exhibited positive skewness**, affecting statistical assumptions.
- **Correlation does not imply causation** - factors such as lender policies, credit history details, and economic conditions may also influence loan approval.
- **Hypothesis testing and regression assume linearity**, which may not fully apply to all variables.
- **Additional relevant factors** (e.g., **debt-to-income ratio, loan purpose**) were not included in the dataset.

Conclusions & Implications

- **Loan-to-income ratio influences loan approval, but there is no strict cutoff rate where loans begin to be rejected.** Instead, **rejection rates gradually decrease after 20-30%, and approval rates increase beyond 30-40%.** This suggests that loan-to-income ratio alone is **not the primary factor** in approval decisions.
- **Higher-income applicants tend to receive higher loan amounts, but income alone is not the most significant approval factor.**
- **Lenders can use these insights to refine approval models, while borrowers should focus on maintaining a balanced loan-to-income ratio and other creditworthiness factors to improve their approval chances.**